

DIRECT ENERGY CONVERSION INTERFACE SPECIFICATION

Six-Channel Hybrid Charged-Particle and Thermal Recovery Edition

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HANDLING	EYES ONLY - AUTHORIZED RECIPIENTS ONLY

1.0 - PURPOSE

This document defines the certified hybrid direct-energy-conversion interfaces serving the six primary A.R.C.-6 reactor cores. The two auxiliary breeder cores use separate isotope-processing and thermal-recovery equipment and are not included in the main DEC dispatch count.

2.0 - PRIMARY ARRAY INTERFACE

- Six independent reactor extraction channels feed a common conditioned DC architecture.
- Each channel can be isolated without disabling the remaining parallel reactors.
- The central A.R.C. Field lattice receives filtered, buffered power only; it is never connected directly to raw converter output.
- Breeder cores B1 and B2 have isolated auxiliary buses and do not inject pulse noise into the Observer power spine.

3.0 - SYSTEM ARCHITECTURE

- Magnetic product-separation nozzles.
- Inductive traveling-wave conversion stages for broad charged-particle spectra.
- Electrostatic deceleration stages for conditioned narrow-band fractions.
- High-voltage DC conditioning, six-channel isolation, and superconducting distribution.
- Thermal recovery for photons, neutrons, neutrals, and unconverted charged-particle power.

4.0 - CERTIFIED PARAMETERS

Parameter	2051 certified range
Directly conditioned exhaust fraction	25-45% of charged-product power
Inductive stage efficiency	58-68% of conditioned fraction
Electrostatic stage efficiency	72-84% of narrow-band fraction
Overall direct-conversion contribution	14-30%, fuel-mode and load dependent
Nominal DC bus voltage	200-800 kV modular
Qualified transient ceiling	1.5 MV with automatic derating above 1.2 MV continuous

5.0 - A.R.C. FIELD POWER QUALITY

The coherence envelope requires exceptionally low phase noise and stable optical timing. Six-channel solid-state conditioning, superconducting DC links, redundant energy stores, and optical isolation prevent reactor or converter transients from disturbing photonic lattice emitters, alignment rings, or the coherence spine.

6.0 - FAULT ISOLATION

- Trip and isolate only the affected primary-core converter channel when possible.
- Maintain the remaining primary reactors in parallel service within certified load-sharing limits.
- Block breeder-core pulse initiation during high-voltage recovery or A.R.C. Field retuning.
- Transfer Observer to reduced operation if filtered power quality cannot sustain the certified coherence envelope.

7.0 - SCIENTIFIC BASIS

The interface is grounded in magnetic-mirror exhaust physics, traveling-wave conversion, electrostatic direct conversion, superconducting DC transmission, high-voltage insulation, and power-electronics isolation. Reactor-scale OSCI demonstrators established the certified ranges used here.

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